Online Content Analysis: Lecture 1

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Text Mining

Course Administration

Grading

- In-class participation, 15%;
- Assignments, 36%, three assignments;
- Final project, 34%;
- Student presentations will be worth 15% of the project.

Attendance

- As by syllabus.

Course Administration

Assignments

- Assignment 1, after lecture 2
- Assignment 2, after lecture 4
- Assignment 3, after lecture 6

Final Project

- Final Report, to be delivered: the day prior to the final presentation, 23h59
- Slides Presentation, to be delivered: the day prior to the final presentation, 23h59
- Final Project Presentation: During the last lecture, 05:00 PM

Dataset

Github repository, Daniel Dan Lab

https://danieldanlab.github.io/

Introduction

What will you learn?

- To use the R/RStudio environment in order to apply Text Mining and Analysis;
- Autonomously gather text information from various sources;
- Discover latent aspects/dimensions in the text through various techniques;
- Label the discovered aspects/dimensions;
- Have a good insight on big volumes of text;
- Blend Text Mining and Marketing;
- Draw conclusions based on the results obtained.
- I hope many more...

Language skills

Homo sapiens is a species that reads and writes.

Reading is the third of the 4 language skills:

- listening;
- speaking;
- reading;
- writing;

(strongly correlated)

Why do we need to analyze online text?

- The user generated content on Social Media platforms produces an impressive quantity of information overload;
- This induces the need for summarization, discovery of latent dimensions in the text, understand the sentiment of the text and induces the necessity to draw conclusions;
- We read on paper and digital screen;
- Huge amounts of text.. it could be too much.

Objective in this lecture and next one

We must manipulate text with the help of R (or other programming languages) to:

- Understand what data, vectors, text strings and characters are;
- Import a text dataset and display strings.
- Obtain information regarding the strings in a dataset.
- Understand what regular expressions are and how they can be useful.
- Use regexp quantifiers, boundaries, common operators, character classes and sequences.

Considerations

It is rare that the (marketing) analyst deals with perfectly well-crafted texts, unless they go several revisions or are written by professionals. Since the primary focus in this book is on analyzing User-Generated Content (UGC) such as online products or service reviews, blog entries or social media posts, we cannot expect any of these texts from being free of linguistic errors, typos and/or misspellings.

Thus, we often encounter situations when we need to search for, select and partly modify or transform words, numbers or whole phrases. This last operation can be called text manipulation and requires specific skills. The main reason to alter text is that we need to create documents that are free of errors, uniform and ready to be analysed.

Why?

Because we are mentally able to correct a misspelled term but computers do not make distinction between a wrongly and a correctly written word: those are just plainly two different words for them. This is important because we need to link the words to numbers in order to draw conclusions and if the text was not polished then the statistics would be flawed.

Some basics

Should we use a spell checker? Are there discrepancies between English language variants? What operations can we employ to manipulate text? Addition, subtraction and substitution.

The Oxford dictionary defines data as: “The quantities, characters, or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.” You can safely see the quantities as numbers, the characters as letters, and the symbols as images and consider all these as data types.

In computer programming we deal with fundamental entities for storing data: variables and constants. Variable: used for storing data assigning an understandable meaningful and concise name to it that can be referenced in order to modify and alter its content based on your needs by other computer routines. Constant: fixed, you cannot alter the value of a constant.

Data, to be easily manipulated, need to be structured and can take the simple form of a vector. How? For example using R...

Short Introduction to R

What is R?

- Is a programming language (like, C, C++, Python, JAVA).

A programming language is a language humans invented to be able to talk to computers in a human-like comprehensible way. It is not as fluent as English but it looks like it. It contains some expressions that will allow you to give commands through simple words, as you do with a pet. With a pet you speak in a simple way, don't you? But with a computer you need to say more than SIT and FETCH. You can give commands to do things that computer are good at: computations and much more.

- Is a language build to find a solution to a problem.
- Is a also (well, better said, it HAS) a programming environment.

What is a programming environment? Everything that helps you host that language (the computer, operating system), run it (compilers), modify what you've written (through a text processor or editor, yes, something like MS WORD but maybe simpler), and test it, is considered programming environment.

In our case, RStudio, is an *Integrated Development Environment* (or plain editor, also shortened as IDE) and the *operating system* itself, both form the programming environment.

More about R

- **R** was created by **R**oss Ihaka and **R**obert Gentleman at the University of Auckland, New Zealand, and is currently developed by the R Development Core Team;
- **R** is an evolution of a commercial software called S. The evolution was... S1, S2, S3, S4, **R** free!
- **R** is a good neighbor, it allows you to integrate its code with procedures written in the C, C++, .Net, Python or FORTRAN so it interacts well with other languages;
- **R** is a specially designed for statistical analysis, graphics representation and reporting;
- **R** is freely available (I love it!) and it is available for various operating systems such as Linux, Windows and Mac (actually and more.)

R Features

- **R** is a well-developed, simple (I think it is simple) and highly effective programming language which includes:

  a. **Conditional propositions** such as: **if** this is valid, **do** that, **if** this **not**, **do** other things;
  b. **Loops**, such as, I like chocolate, repeat, I like chocolate, repeat. How many times? 100? 1000? You choose, depends on your ~~mood~~ needs;
  c. User defined **recursive functions**;
  d. **Input** and **output** facilities. **Input**: I have some **data**. **Output**: after I *manipulate* them, I present a *table*, a *graph* or I *print* them, etc...

- **R** has an effective data handling so that I can manipulate them a lot, it is a sort of a game, and storage facility where to save data (yes, I have to put data in a safe place otherwise I might not find it again).

R Data Storage

R provides a suite of operators for calculations on

a. **Arrays**. An array is a place to store related items.

85	90	68				29	71

b. **Vectors**. A **vector** is just an array wrapped in a type.

Length	Vector	Type
1	1	Integer
2	3.5 8.2	Numeric
3	TRUE FALSE TRUE	Logical
4	'John' 'Jack' 'Jill' 'Jim'	Character

c. **Lists**. A **list** is a special type of array used to save elements of different types.

```
> x <-list(1,'a',TRUE, 1+4i);
> x
[[1]]
[1] 1

[[2]]
[1] "a"

[[3]]
[1] TRUE

[[4]]
[1] 1+4i
```

d. **Matrices**. A **matrix** is a 2D **array**.

2x2 matrix	column 1	column 2
row 1	1	2
row 2	3	4

More with R

- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.
- R is world's most widely used statistics programming language! It's the # 1 choice of data scientists in academia and not only and supported by many important contributors.
- R is taught in universities, but it is deployed in industry as well as in mission critical business applications.

What businesses use R?

- Facebook - For behavior analysis related to status updates and profile pictures.
- Google - For advertising effectiveness and economic forecasting.
- Twitter - For data visualization and semantic clustering
- Microsoft - Acquired Revolution R company and use it for a variety of purposes.
- Uber - For statistical analysis
- Airbnb - Scale data science.
- IBM - Joined R Consortium Group
- ANZ - For credit risk modeling

Click here for a list of companies using R.

Get R/RStudio

Two step install: 1. R, 2. RStudio

1. **Install R**, get the latest version

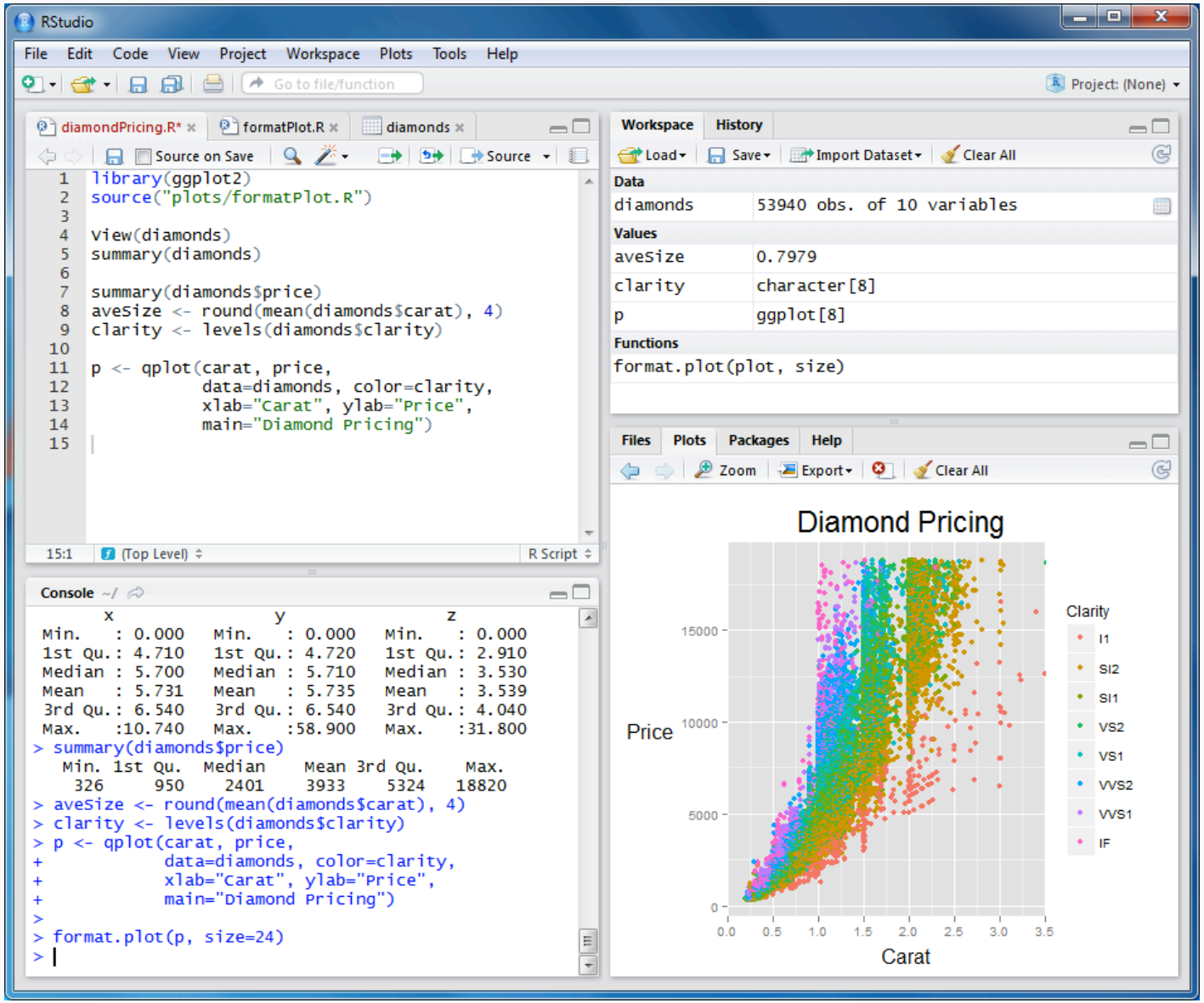
   - **Windows**: https://cran.r-project.org/bin/windows/base/
   - **OSX**: https://cran.r-project.org/bin/macosx/

2. **Install RStudio**: https://posit.co/download/rstudio-desktop/

   - **RStudio Desktop Free**
     - Windows 10/11 Install the .EXE
     - macOS 11+ Install the .DMG

Alternatively, use the cloud version: https://posit.cloud

RStudio



Code Editor, R Console, Workspace and history, Plots and files, all in one program

Important!

1. What is your working directory?

- Commands getwd() and setwd(). The first one stands for get working directory, the second for setting the working directory.
- You can select a folder and set it as your working directory: **Session** > *Set your Directory*

2. How to install packages?

- Use the function **install.packages**("packagename")

3. How to load a library (from a package)

- Use the function **library**(packagename)

4. How to import data from comma separated values files, csv:

```
variable <- read.csv("myfile.csv", header = TRUE, sep = ",")
```

5. How to ask for help

- In the RStudio console: **?read.csv**

More on Data Types

A vector is one of the data structures R frequently uses, alongside with: lists, matrices, data frames, tables and factors.

A vector can be atomic and contains a single value, or many, also called array, which contains more than one value.

The difference between a list of vectors and a data structure list is that a list of vectors can contain only a single data type while a data structure list can contain different types of data, even lists of lists.

The **R** language fits data into six distinctive vector types: three number types (integer, numeric, complex), a logical type (True or False), a character type and a raw type.

Text strings belong to the character class.

One can see a variable as a drawer that contains information. In the same way we can set up a drawer to contain socks in but for now we leave it empty, an empty string *empty_string* can be assigned to a variable, i.e. a string containing no characters (we will see that). Paradoxically this will still belong to the character class. When we are unsure of the class type of a variable we can check the type with the function **class**().

Example

Exercise: Get the details of the variables (objects) in memory

```
ssstring <- "Snake"
numberint1 <- 10
numbernum2 <- 0.5
arrayofintegers <- c(10:60, by=10)
print(ls())
print("Details of the objects in memory:")
print(ls.str())
```

What do you think is the difference between a number and an integer?

Note, **ls** stands for **l**ist and **str** stands for **str**ucture.

```r
(empty_string <- "")
```

```
## [1] ""
```

```r
class(empty_string)
```

```
## [1] "character"
```

Similarly, we can create an empty character by using the function **character**() setting the length argument (the parameters inside the parenthesis of a function) to zero.

```r
empty_character <- character(length = 0)
empty_character
```

```
## character(0)
```

```r
class(empty_character)
```

```
## [1] "character"
```

What if we wanted to have a vector of character type but containing no elements in it? We can see this as character type vector initialization.

```r
character_vector <- character(10)
character_vector
```

```
##  [1] "" "" "" "" "" "" "" "" "" ""
```

We can assign words or strings to this newly created empty vector by selecting the vector's position we want to occupy. In our case, we want to occupy the fifth position of the vector with the word "five" and we access it by indexing. Indexing is achieved by adding square brackets after the vector's name and insert the position which is given by a number.

```
character_vector[5] <- "fifth"
character_vector
```

```
##  [1] ""      ""      ""      ""      "fifth" ""      ""      ""      ""
## [10] ""
```

One must be fully aware that a vector is for homogeneous storage, meaning that, it can only hold data of the same type. If we had a number type vector and we append a string to it, R will convert the numeric values it holds inside the vector into strings.

If we need full control over the vectors we can convert the vector types ourselves. We can check if a vector is of character type with the help of the function **is.character**().

Casting from one type to another, for example from numeric to character, can be accomplished through the function **as.character**() because we might want to have numbers written as a text. Number types are used for computations while numbers written as a text belong to the character type and can be interpreted as words.

Some Basics

In the following example we can see two variables, **x** and **y** holding two different objects: character and numeric, respectively. We can check specifically if they are of character type or not. Use the **class**() type and cast one type into another and check again to see if the last operation successfully occurred.

```
x <- "really, is this a character?"
y <- 9+8

is.character(x)
```

```
## [1] TRUE
```

```
is.character(y)
```

```
## [1] FALSE
```

```
class(x)
```

```
## [1] "character"
```

```
class(y)
```

```
## [1] "numeric"
```

```
y <- as.character(y)
class(y)
```

```
## [1] "character"
```

There are situations in which we need to put together different strings to create longer ones by concatenating them. The **paste**() function will accomplish this. It has two arguments: *sep* and *collapse*.

```
x <- paste("My favorite book", "has many chapters", sep = " ")
x
```

```
## [1] "My favorite book has many chapters"
```

```
chapters <- paste("Chapter", 1:5, sep=".", collapse = ", ")
chapters
```

```
## [1] "Chapter.1, Chapter.2, Chapter.3, Chapter.4, Chapter.5"
```

Perhaps you have noticed that we used the same variable, **x**, as in the previous example. Whenever you reuse the name of a variable you overwrite it with the new data, so it is a good idea to keep good track of the names you have already given.

Working with strings

In text mining, strings are our bread and butter and for a better comprehension, let us make some simple text manipulation examples. We import the "Product Reviews.csv" dataset select the *Title* column and display a small set (20 items) to give an idea of how the data looks like. The data are selected randomly, but with the aid of the **set.seed**() function the reproducibility of the results is assured and randomness is there but kept in a cage.

```
options(stringsAsFactors = FALSE)
prodreviews <- read.csv("Product Reviews.csv")
prodtitle <- as.character(prodreviews$Title)
set.seed(123)
(prodtitle20 <- sample(prodtitle, 20))
```

Working with strings

As we can see, the system displays them in a reproducible (pseudo) random sequence. Note that text is case sensitive and for the computer the words “great” and “Great” are two different entities. Just to warm up a bit, let us select a part of the strings with the help of the function **substr**(). Our goal could be, for example, to select the first four letters of the title string. Technically speaking, we select the portion of the name starting from the index position of the first letter of a word, and ending with a desired position, in our case the fourth.

```
substr(prodtitle20, start = 1, stop = 4)
```

We notice that we have abbreviated the names and some strings have identical sequences which are made by a combination of lowercase or uppercase letters, spaces and punctuation marks. Since we might not want to make confusion by having duplicates, we can resort to something different. A useful command in cases like this could be **abbreviate**(). It will pick a minimum of 4 letters from each string in the data set in such a way so that they remain unique after its application. Generally, the command will select the first letter of all the words in the string. If the words are less than four then it will construct other abbreviations. In any case the results will be crafted in order to obtain double occurrences. You need to abbreviate for several purposes such as to save time and (computer) space.

abbreviate(prodtitle20)

Working with strings

You will find two abbreviations to be duplicated, namely, "KndF", this is because the original string was the same. The length of the strings is a statistic datum and if we wonder how many characters each string of the sample has, or which is the string with the greatest number of letters. For this purpose, we may use the functions **nchar**() and **max**() applied to the $prodtitle20$ data set. This is an important start as a marketing analyst for texts: how much do people talk about a product?

```
nchar(prodtitle20)
max(nchar(prodtitle20))
```

We obtain a vector of numbers representing how many characters each title is composed of and another one containing the maximum length from all the titles. In the latter case, the next step is to find to what string that number corresponds. We can use the function which to detect the string(s) with the longest name. As such, we obtain two strings of 49 characters each because it happens that two strings have the same (long) length.

Working with strings

```
prodtitle20[which(nchar(prodtitle20)==max(nchar(prodtitle20)))]
```

In plain English, this can be translated from **R**: which is the string that has the maximum number of characters inside the sample data set *prodtitle20*? How can we find how many titles are in the data set and how many of these titles are unique? We can use the functions **length**() and **unique**() to give an answer to both previous questions.

```
length(prodtitle)
length(unique(prodtitle))
```

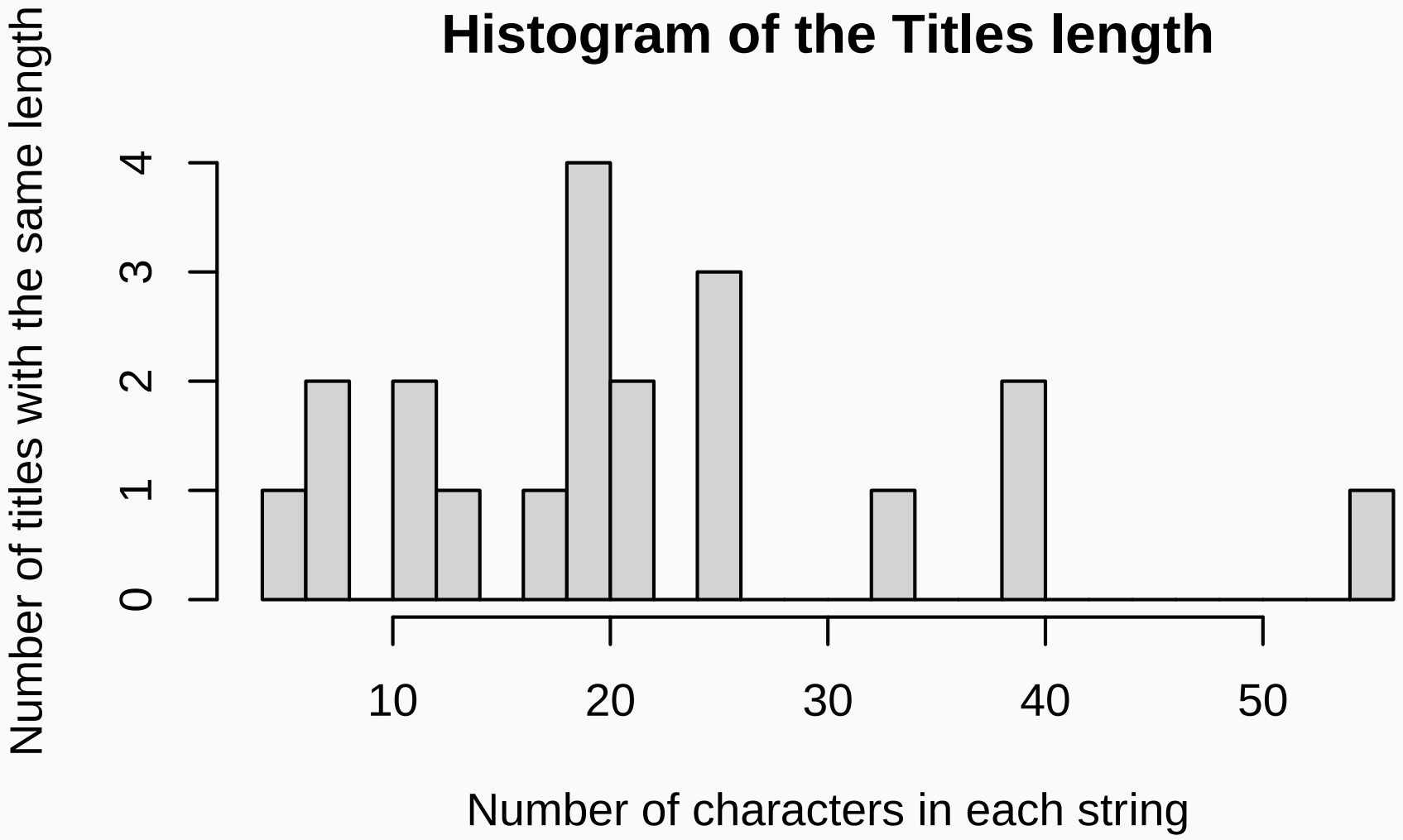
Because we have mentioned the word statistics applied to this data set, a quick overview can be obtained with the help of the function **summary**() regarding the number of characters of each string and as such we shift our attention to the whole *prodtitle* data set.

```
summary(nchar(prodtitle))
```

Working with strings

But how are these string lengths distributed, namely how many strings have which lengths?

```
hist(nchar(prodtitle20), main = 'Histogram of the Titles length',
breaks = 20,
xlab = 'Number of characters in each string',
ylab = 'Number of titles with the same length')
```



What can we notice from the previous histogram?

End of Lecture 1

That was all for today! Questions? :)